

HINTS AND SOLUTIONS

Q.1 A

Sol. $\frac{1}{f_l} (1.5 - 1) \left[\frac{1}{30} - \frac{1}{30} \right]$
 $\Rightarrow f_l = 30 \text{ cm}$

$$P = P_1 + P_2 + P_3 = \frac{1}{30} + \frac{1}{30} + \frac{1}{15} = \frac{4}{30} = -\frac{1}{F_m}$$

$$\therefore F_m = -7.5 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{-60} = -\frac{2}{15} \Rightarrow \frac{1}{v} = \frac{1}{60} - \frac{2}{15} \Rightarrow v = -60/7$$

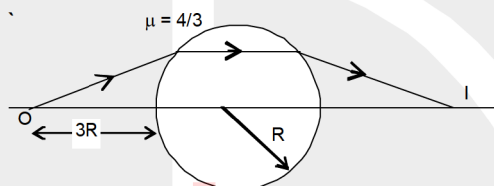
Q.2 C

Sol. By relation,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{4}{3v} - \frac{1}{-3R} = \frac{4/3 - 1}{R}$$

$$\text{or } \frac{4}{3v} = 0 \Rightarrow v = \infty$$



Q.3 B

Sol. Refraction of flat surface:

Apparent depth = (Real depth) / (R.I.)

$$\Rightarrow t_1 = \frac{t}{n} \quad \dots\dots(i)$$

Refraction of curved surface

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

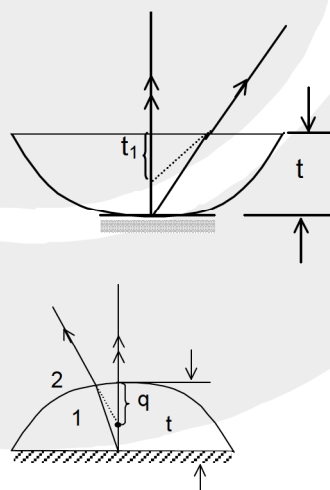
Here $n_1 = n$, $n_2 = 1$, $u \cong -t$, R is $-R$

$$\Rightarrow \frac{1}{v} - \frac{n}{-t} = \frac{1-n}{-R}$$

$$\Rightarrow \frac{1}{v} = \frac{n-1}{R} - \frac{n}{t}$$

Putting $v = t_2$ we obtain, $-\frac{1}{t_2} = \frac{n-1}{R} - \frac{n}{t} \quad \dots\dots(2)$

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{ Putting } R_1 = R \text{ and } R_2 = \infty \text{ we obtain}$$



$$f = \frac{R}{n-1} \quad \dots\dots(3)$$

Putting $\frac{n-1}{R} = \frac{1}{f}$ from (3) in (2) we obtain,

$$-\frac{1}{t_2} = \frac{1}{f} - \frac{n}{t} \quad \Rightarrow \quad \frac{1}{f} = \frac{n}{t} - \frac{1}{t_2}$$

$$\Rightarrow f = \frac{tt_2}{nt_2 - t}$$

Putting the value of n from equation (1)

$$\Rightarrow f = \frac{t_1 t_2}{\frac{t}{t_1} t_2 - t} = \frac{tt_1 t_2}{tt_2 - tt_1} = 75 \text{ cm}$$

Q.4 D

Q.5 C

Sol. Magnification remains the same but brightness reduces to 1/4.

Q.6 A

Sol. $\frac{\mu_1}{x} - \frac{\mu_2}{-x} = \frac{\mu_1 - \mu_2}{R}$

$$\Rightarrow \frac{\mu_1 + \mu_2}{x} = \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow x = \frac{R(\mu_1 + \mu_2)}{(\mu_1 - \mu_2)}$$

Since x is positive, $\mu_1 > \mu_2$.

Q.7 B

Q.8 A

Sol. Apparent depth = $\frac{t_1}{\mu_1} + \frac{t_2}{\mu_2}$

$$\text{Shift} = (t_1 + t_2) - \left(\frac{t_1}{\mu_1} + \frac{t_2}{\mu_2} \right)$$

$$3.0 = (t_1 + 10) - \left(\frac{t_1}{1.5} + \frac{10 \times 3}{4} \right)$$

$$3.0 = \frac{t_1}{3} + 2.5$$

$$t_1 = 1.5 \text{ cm}$$

Q.9 A

Sol. $x = 2y$

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{y} - \frac{1}{2y} = \frac{1}{f}$$

$$\frac{f}{y} = 2$$

Q.10 D

Sol. $\frac{1}{10} = (1.5 - 1) \left(\frac{1}{R} + \frac{1}{R} \right)$

$$\frac{1}{10} = 0.5 \times \frac{2}{R} \Rightarrow R = 10 \text{ cm}$$

Refraction from 1st surface, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\frac{1.5}{v_1} - \frac{1}{-20} = \frac{1.5 - 1}{+10} \Rightarrow v_1 = \infty$$

for the second surface,

$$\frac{2}{v} - \frac{1.5}{\infty} = \frac{2 - 1.5}{-10}$$

$$v = -40 \text{ cm}$$

Q.11 B

Sol. For plane mirror, $u = v = x$ (let)

For lens,

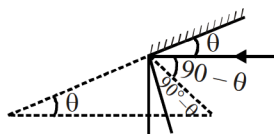
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{x} - \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \frac{2}{x} = \frac{1}{f}$$

$$\Rightarrow x = 2f$$

Q.12 B

Sol.



$$\theta - (90 - \theta) = 2\theta - 90^\circ$$

$$\tan(2\theta - 90^\circ) = \frac{x}{r}$$

$$-\cot 2\theta = \frac{x}{r} \Rightarrow \frac{2 \tan 2\theta}{1 - \tan^2 \theta} = \frac{-r}{x}$$

$$\frac{2 \times \frac{r}{h}}{1} = \frac{r}{x} \Rightarrow x = \frac{h}{2}$$

Q.13 D

Sol. Assertion and Reasoning both are incorrect.

Q.14 D

Sol. $n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$

n increase λ decrease

Statement-2 : Cauchy's relation is for same medium and different wave length.

Statement-1 : Snell's law for same wavelength is different medium.

Q.15 C

Sol. Focal length will be different for the 2 cases. Also we can't apply the formula of thin lens here.

Q.16 BC

Sol. $a + b = D = 180$

$$|m| = \frac{b}{a} = 2 \Rightarrow b = 2a$$

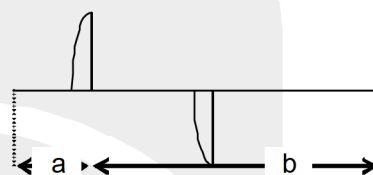
which gives $a = 60$ cm, $b = 120$ cm

separation between two lenses $= b - a = 60$ cm

from $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\Rightarrow \frac{1}{f} = \frac{1}{+120} + \frac{1}{60} = +\frac{3}{120} = \frac{1}{40}$$

$f = 40$ cm



Q.17 AB

Sol. For $\mu_2 > \sqrt{2}$ (TIR will not take place)

$$2 \sin 45^\circ = \mu_2 \sin r$$

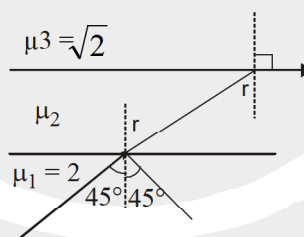
$$\mu_2 \sin r = \sqrt{2} \sin e$$

$$e = 90^\circ$$

Hence deviation is 45°

$\mu_2 < \sqrt{2}$ TIR will take place

deviation is 90° .



Q.18 ABD

Sol. From given data, using

$$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}, \text{ we get}$$

$$f_A = -\frac{4}{3} \text{ cm}, \quad f_B = -8 \text{ cm}$$

$$f_C = -3 \text{ cm}, \quad f_D = -\frac{3}{2} \text{ cm}$$



Q.19 [Ans. 0010]

Sol. $NS = x \left(1 - \frac{1}{4/3} \right) = \frac{x}{4} \text{ cm}$

obj. at $5 + x - \frac{x}{4} = 5 + \frac{3x}{4}$

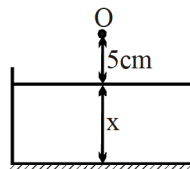


Image at $5 + \frac{3x}{4}$ behind mirror.

Final image = $5 + x - \frac{x}{4} + 5 + \frac{3x}{4} = 10 + \frac{3x}{2} = 25 \text{ cm}$

$x = 10 \text{ cm}$]

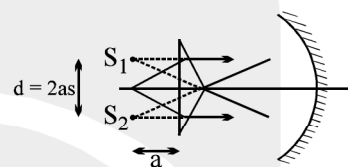
Q.20 [Ans. $\pi/20 \text{ cm}$]

Sol. Object size = $d = 2 \times 10 \times \left(\frac{3}{2} - 1 \right) \times \frac{1.8 \times \pi}{(180)} = \frac{\pi}{10} \text{ cm}$

$v = \frac{uf}{u-f}$

$m = \frac{f}{u-f} = \frac{10}{-30 - (-10)} = -\frac{1}{2}$

$I = (m) d = -\pi/20$]



Q.21 [Ans. 0.27 I]

Sol. $u = -20, f = -20$

gives, $v = -10$

Let P = Power of source

$I = \frac{P}{4\pi(60)^2} \dots(1)$

Energy received by lens

$E_2 = \frac{P}{4\pi(20)^2} A_1$

$\therefore I_2 = \frac{0.75E_2}{A_2} \dots(2)$

From similar triangles

$\frac{A_2}{A_1} = 25$

$\therefore I_2 = 0.27 I$

